
GROUP 3 MODULE III PROJECT

Movie Studio Production Analysis

GROUP MEMBERS

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PROJECT DEFENSE

1. Business Context and Objectives

Turning historical movie data into practical decision support.

- Our company plans to enter the film industry by launching a new movie studio. However, with limited experience in movie production, the company needs data-driven insights to guide its initial investment and production decisions to ensure commercial success.
- Key decisions include:
 - **Which movie genre we should invest in?**
Identify genres with strong financial returns.
 - **What is the best time of year to release a movie?**
Identify the strongest periods of the year for box-office performance
 - **Does production budget relate to movie performance?**
Understand the relationship between production spending and box-office success.
- To support these decisions, we analyzed historical movie and box-office data to identify patterns associated with stronger commercial performance.

2. Data Sources & Project Structure

Financial files + IMDb SQLite data + documented project workflow.

- We analyzed historical movie data from :
 - IMDb
 - Box Office Mojo
 - TMDB
 - Rotten Tomatoes
 - The Numbers.
- We combined information on:
 -  **Movies & Genres**
to understand which types of films generated stronger returns.
 -  **Release Dates**
to identify periods associated with stronger box-office performance.
 -  **Production Budgets and Revenue**
to examine how investment levels relate to box-office results.
- The analysis focused on turning historical movie data into practical insights that could support the studio's production and investment decisions.

3. Data Preparation & Feature Engineering

Preparing raw data for reliable analysis.

Removed '\$' and commas from monetary fields and converted them to numeric values.

Converted `release_date` to datetime and extracted `release_month`.

Calculated `net_profit = worldwide_gross - production_budget`.

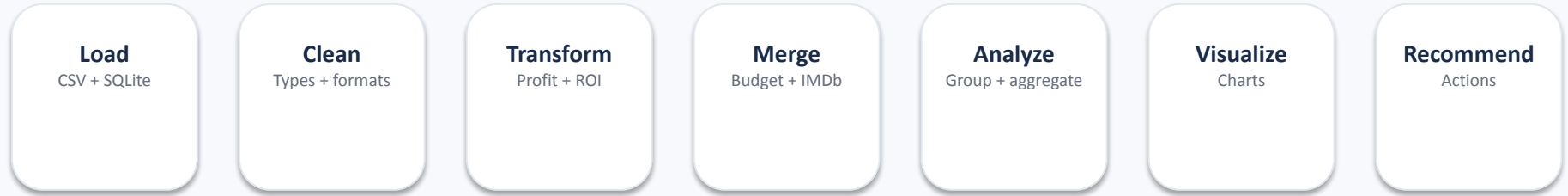
Calculated `ROI = (net_profit / production_budget) × 100`.

Created `loss_or_profit` to classify movies as Profit or Loss.

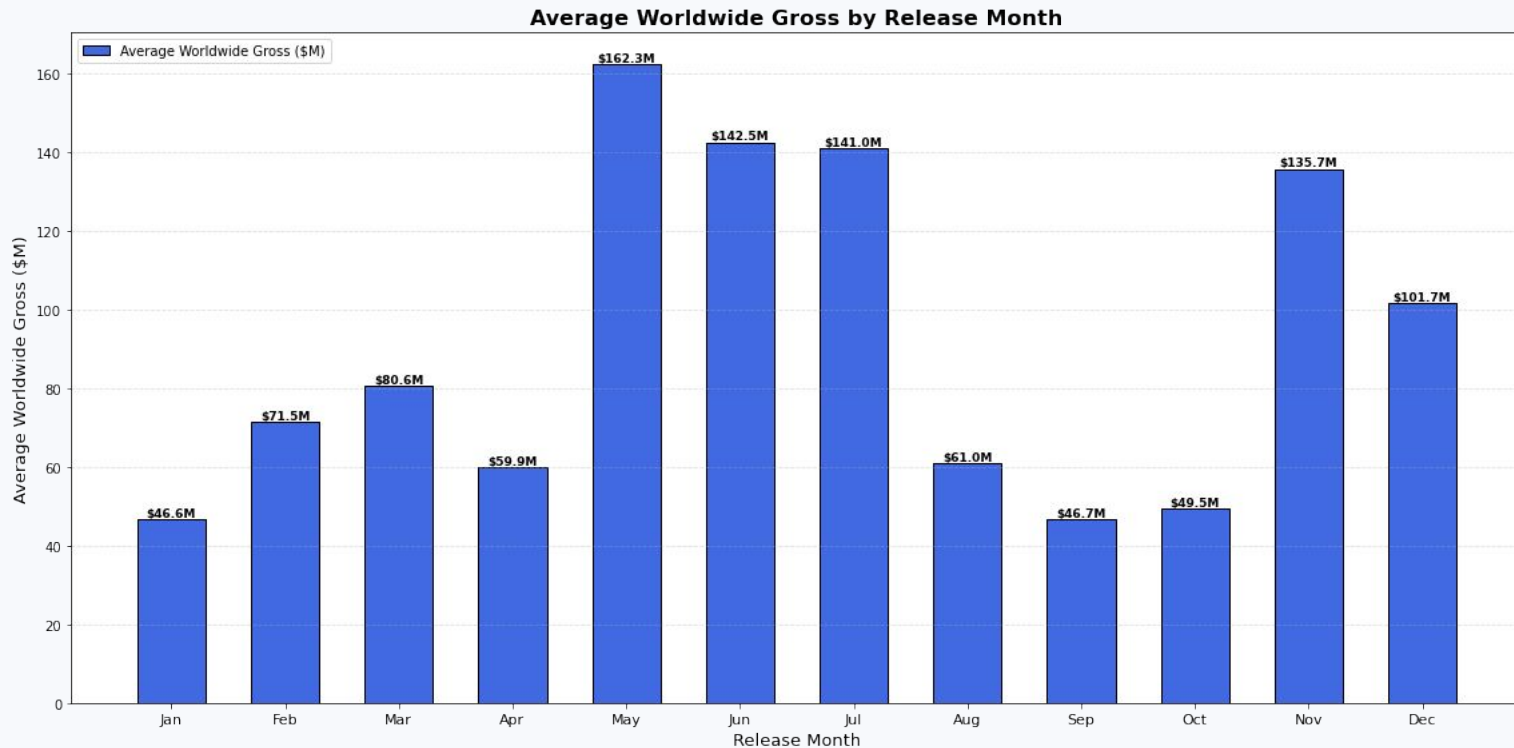
Split multi-genre strings and used `explode()` to analyze individual genres.

4. Analytical Workflow

From raw data to business recommendations.



Business Question 1: Which Release Months Perform Best?



Insight: Best release window is May-July and November-December.
May has the highest average worldwide gross at about \$162.3M.
June and July follow at about \$142.5M and \$141.0M.

Business Question 1: Conclusion & Recommendation

Conclusion

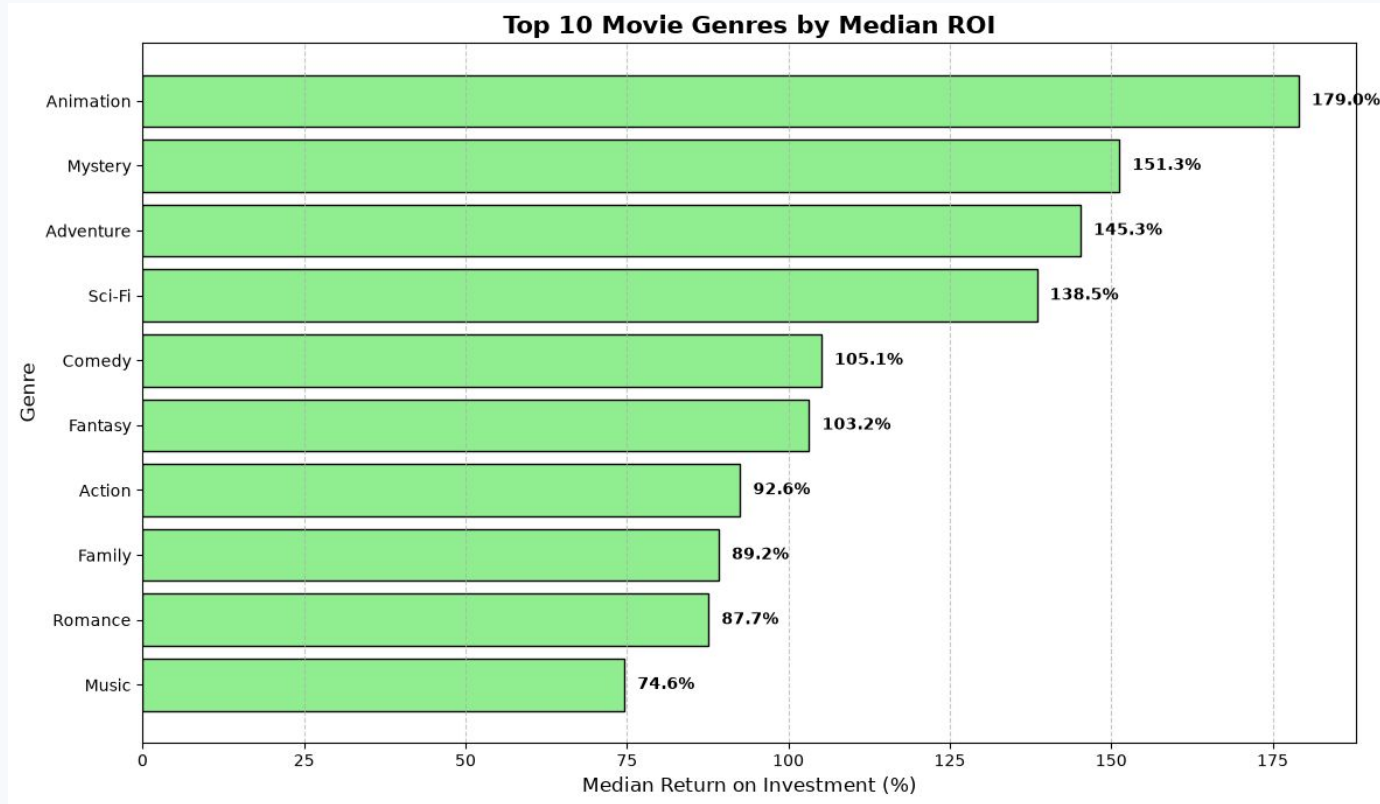
May–July and November–December emerge as prime release windows. May is the strongest individual month by average worldwide gross.

Recommendation

Prioritize commercially strong releases in these windows while considering competition, genre, marketing, audience behavior and release volume.

Business Question 2: Which Genres Have the Highest ROI?

Top 10 genres ranked by median ROI.



Business Question 2: Conclusion & Recommendation

Conclusion

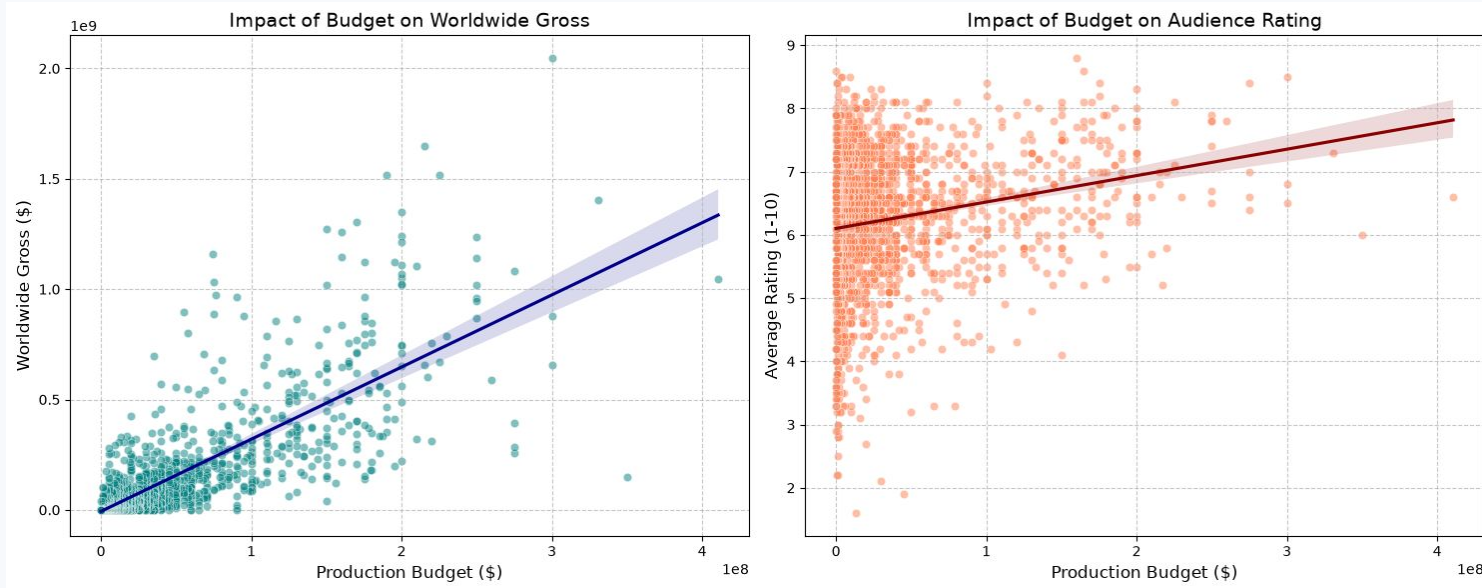
Animation leads with a 179.0% median ROI, followed by Mystery (151.3%), Adventure (145.3%) and Sci-Fi (138.5%).

Recommendation

Give greater consideration to high-ROI genres, but combine genre with budget, target audience, competition, release timing and marketing strategy.

Business Question 3: How Does Production Budget Affect Performance?

Budget is compared with worldwide gross and audience rating.



Higher production budgets generally correspond with higher worldwide gross.

The relationship is not perfect: similar budgets can produce very different revenues.

Production budget shows a relatively weak relationship with audience rating.

Business Question 3: Conclusion & Recommendation

Conclusion

Higher budgets are generally associated with higher worldwide gross, but budget size alone does not guarantee commercial success or higher audience ratings.

Recommendation

Optimize production budgets using expected revenue, historical ROI, genre performance, release timing, target audience, competition and marketing requirements.

Overall Findings

The three business questions form one decision framework.

WHEN?

May–July and November–December are the prime release windows identified by historical average worldwide gross.

WHAT?

Animation, Mystery, Adventure and Sci-Fi show the strongest median ROI among the top 10 genres.

HOW MUCH?

Optimize production budgets based on expected return and risk rather than assuming bigger is always better.

Genre → Budget → Expected ROI → Release Window → Marketing Strategy → Investment Decision

Limitations & Future Work

What the analysis can—and cannot—tell us.

Historical relationships do not guarantee future box-office outcomes.

ROI is based on worldwide gross and production budget; additional costs such as marketing are not included.

Future work could incorporate marketing spend, competition, franchise strength, distribution strategy and audience demographics.

Predictive modeling could estimate expected gross or probability of profitability for future releases.

Final Takeaway

Is Movie profitability and performance influenced by:

- Release Timing? 
- Genre? 
- Production Budget?

Thank You | Questions & Discussion